

Describing Data and Distributions

CENTER, SPREAD, AND THE MATHEMATICS OF VARIATION

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Today we put numbers on what we see

Last time, we used visualizations to identify patterns in data.

Today we ask:

- ▶ Where is the **center** of a distribution?
- ▶ How much **variation** is there?
- ▶ How do extreme observations affect our summaries?
- ▶ How can we translate a graph into a defensible numerical description?

Visualization $\longrightarrow$ Calculation $\longrightarrow$ Interpretation

A numerical distribution: four things to describe

1. Shape

Symmetric? Skewed? One peak or several?

2. Center

What value represents a typical observation?

3. Spread

How much do observations vary?

4. Unusual observations

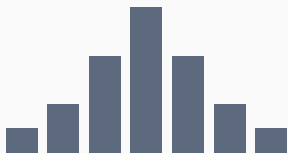
Are there values that stand apart from the rest?

This is the same backbone we will use whenever we describe a quantitative variable.

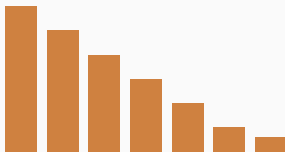
Start with the picture.

Then decide which numbers make sense.

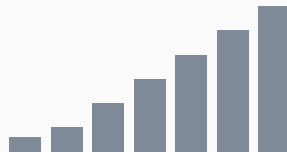
Shape: symmetry, skewness, and modality



roughly symmetric



right-skewed



left-skewed

- ▶ **Skew** is named for the direction of the long tail.
- ▶ **Modality** describes prominent peaks: unimodal, bimodal, and so on.

Your turn: predict the shape before seeing data

Your Turn

What shape would you expect for each variable? Explain why.

1. Household income in a large U.S. city
2. Scores on an easy exam
3. Adult height within a single sex category
4. Number of emergency-room visits in a year

Think about natural lower or upper bounds and the possibility of a few extreme values.

What is a “typical” observation?

The sample mean

For a sample of n observations, the mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

In words:

$$\text{mean} = \frac{\text{sum of all observed values}}{\text{number of observations}}$$

Notation matters

$\bar{x}$ = sample mean μ = population mean

Our turn: calculate a mean

Our Turn

Suppose five neighborhoods have an illustrative policy score of

2, 4, 5, 7, 12.

What is the sample mean?

$$\bar{x} = \frac{2 + 4 + 5 + 7 + 12}{5} = \frac{30}{5} = \boxed{6}$$

The mean uses every observation in the dataset.

The mean is a balance point

Using the same data, $\bar{x} = 6$.

x_i	2	4	5	7	12	
$x_i - \bar{x}$	-4	-2	-1	1	6	0

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

Important

Positive and negative deviations from the mean exactly cancel. This will matter when we measure spread.

The median

The **median** is the middle value after sorting observations from smallest to largest.

Odd number of observations

2, 4, 5, 7, 12

Median = 5.

Even number of observations

2, 4, 5, 7, 12, 14

Median = $(5 + 7)/2 = 6$.

The median depends on order, not the exact distance between values.

Mean vs. median: add one extreme value

Your Turn

How do the mean and median change?

Dataset A

30, 50, 70, 90

$\bar{x} = 60$, median = 60

Dataset B

30, 50, 70, 1000

$\bar{x} = 287.5$, median = 60

The mean moves dramatically. The median does not.

Why policy analysts often care about the median

Imagine five illustrative home values, in thousands of dollars:

250, 275, 300, 325, 1200

Mean

$$\bar{x} = 470$$

\$470,000

Median

$$\tilde{x} = 300$$

\$300,000

Your Turn

Which number better describes a “typical” home in this tiny example?

What information does the other number still convey?

Center is not enough.

Same center, very different spread

Dataset A

4, 5, 6, 7, 8

Mean = 6

Dataset B

0, 3, 6, 9, 12

Mean = 6

Same center. Different variation.

A useful description of a quantitative variable usually needs both center *and* spread.

The range

The simplest measure of spread is

$$\text{Range} = \max(x) - \min(x)$$

For

2, 4, 5, 7, 12,

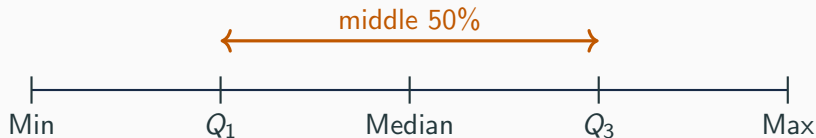
$$\text{Range} = 12 - 2 = 10.$$

Limitation

The range depends only on the two most extreme observations.

Quartiles and the interquartile range

Quartiles divide ordered data into four parts.



$$IQR = Q_3 - Q_1$$

The IQR measures the spread of the middle 50% of observations.

Our turn: quartiles and IQR

Use the **median-of-halves** convention for today.

2, 4, 5, 7, 8, 10, 12, 14

$$\text{Median} = \frac{7 + 8}{2} = 7.5$$

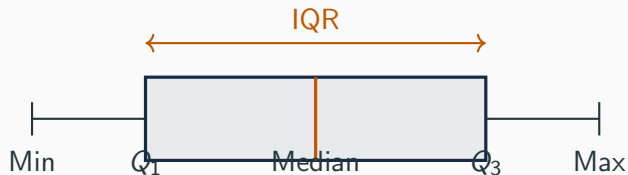
$$Q_1 = \frac{4 + 5}{2} = 4.5$$

$$Q_3 = \frac{10 + 12}{2} = 11$$

$$IQR = 11 - 4.5 = \boxed{6.5}$$

Different software can use slightly different quartile conventions for some datasets. The underlying idea is the same.

A boxplot compresses the distribution



A boxplot makes it easy to compare **center**, **spread**, **skew**, and possible **outliers**.

The $1.5 \times \text{IQR}$ rule for suspected outliers

A common rule flags observations outside these fences:

$$\text{Lower fence} = Q_1 - 1.5(\text{IQR})$$

$$\text{Upper fence} = Q_3 + 1.5(\text{IQR})$$

An observation beyond a fence is a **suspected outlier**.

Do not automatically delete it.

An unusual value may be an error, but it may also be the most substantively important case in the dataset.

Our turn: is 30 an outlier?

Consider

2, 4, 5, 7, 8, 10, 12, 30.

Using the median-of-halves convention:

$$Q_1 = 4.5, \quad Q_3 = 11, \quad IQR = 6.5.$$

$$\text{Lower fence} = 4.5 - 1.5(6.5) = -5.25$$

$$\text{Upper fence} = 11 + 1.5(6.5) = 20.75$$

$$30 > 20.75 \quad \Rightarrow \quad 30 \text{ is a suspected outlier.}$$

How far are observations from the mean?

Why not just average the deviations?

For

$$2, 4, 5, 7, 12, \quad \bar{x} = 6,$$

we have:

x_i		2	4	5	7	12
$x_i - \bar{x}$		-4	-2	-1	1	6

$$-4 - 2 - 1 + 1 + 6 = 0.$$

The deviations always sum to zero, so their ordinary average cannot measure spread.

The solution: square the deviations

Squaring does two useful things:

1. removes negative signs;
2. gives larger deviations more weight.

$x_i - \bar{x}$	-4	-2	-1	1	6
$(x_i - \bar{x})^2$	16	4	1	1	36

$$\sum (x_i - \bar{x})^2 = 58.$$

Sample variance

The sample variance is the average squared distance from the mean, with an adjustment for estimating the mean from the same sample:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

For our five observations:

$$s^2 = \frac{58}{5 - 1} = \frac{58}{4} = 14.5.$$

Units

If x is measured in points, variance is measured in *points squared*.

Standard deviation brings us back to the original units

Take the square root of the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

For our example:

$$s = \sqrt{14.5} \approx \boxed{3.81}.$$

Interpretation

The standard deviation summarizes the typical scale of distance between observations and the mean. Smaller s means values are more tightly clustered; larger s means more variation.

Put the entire calculation together

For 2, 4, 5, 7, 12, $\bar{x} = 6$.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	-4	16
4	-2	4
5	-1	1
7	1	1
12	6	36
Sum = 0		Sum = 58

$$s^2 = 58/4 = 14.5, \quad s = \sqrt{14.5} = 3.81.$$

Same mean, different standard deviation

Dataset A

4, 5, 6, 7, 8

$$\bar{x} = 6, \quad s \approx 1.58$$

Dataset B

0, 3, 6, 9, 12

$$\bar{x} = 6, \quad s \approx 4.74$$

Your Turn

What does the larger standard deviation tell us substantively about Dataset B?

Why do we divide by $n - 1$?

When we calculate s^2 , we already used the sample to estimate $\bar{x}$.

Once $n - 1$ deviations are known, the final deviation is forced because

$$\sum (x_i - \bar{x}) = 0.$$

That leaves $n - 1$ degrees of freedom.

For now

You should understand the intuition and recognize the formula. We will return to degrees of freedom when we study statistical inference.

Robust and non-robust statistics

	Center	Spread
Sensitive to extremes	Mean	Standard deviation
More robust	Median	IQR

- ▶ Roughly symmetric distribution, no major outliers: **mean + SD**
- ▶ Strongly skewed distribution or important outliers: **median + IQR**

Look at the distribution *before* choosing the summary.

Your turn: what would you report?

Your Turn

Choose a center and spread for each case, and explain your choice.

1. Household income with a long right tail
2. Adult heights with an approximately symmetric distribution
3. Emergency-room wait times with several extremely long waits
4. Scores on an exam with no strong skew or outliers

Do not choose based on habit. Choose based on what the distribution looks like.

From hand calculation to R

You should understand what the statistics mean before asking software to compute them.

Common R functions

`mean(x)`

`median(x)`

`range(x)`

`IQR(x)`

`var(x)`

`sd(x)`

`summary(x)`

R does the arithmetic. You still decide what to calculate and how to interpret it.

A complete descriptive statement

Suppose a distribution is right-skewed with a median of 18 minutes and an IQR of 11 minutes.

A strong description might say:

“Wait times are right-skewed. The median wait is 18 minutes, and the middle 50% of observations span 11 minutes. A small number of substantially longer waits produce the right tail.”

Notice

Shape + center + spread + unusual observations = a substantive description.

Exit check

For the data

3, 4, 5, 6, 7, 8, 23,

work with a partner and answer:

1. What are the mean and median?
2. Which is more representative of a typical value?
3. What is the range?
4. Would you prefer mean + SD or median + IQR? Why?
5. What would you inspect before calling 23 an error?

What I want you to leave with

1. **Start with the distribution.** Describe shape and unusual observations.
2. **Center needs context.** Mean and median answer related but different questions.
3. **Spread matters.** Two datasets can have the same center and very different variation.
4. **Know the mathematics.** Variance and standard deviation are built from deviations from the mean.
5. **Match the statistics to the data.** Mean + SD for roughly symmetric data; median + IQR for skewed data or outliers.

See it. Calculate it. Interpret it.